Requirement Management Plan (RMP)

Team Name:

ProjectMasters

Project Title:

National Freelance and Skill Verification Platform

Module:

Communication Module

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Table of Contents

1. Introduction 2
   1.1 Purpose 2
   1.2 Scope 2
   1.3 Definitions, Acronyms, and Abbreviations 2
2. Requirements Management Approach 2
   2.1 Requirements Planning 2
   2.2 Requirements Tracking 3
   2.3 Requirements Reporting 3
3. Configuration Management 3
   3.1 Version Control 3
   3.2 Change Management Process 3
   3.3 Baseline Management 4
   3.4 Access Control 4
4. Requirements Prioritization 4
5.Requirements Elicitation Techniques 5
6. Product Metrics 5
   6.1 Performance Metrics 5
   6.2 Quality Metrics 5
   6.3 Usage Metrics 5
   6.4 Security Metrics 5
7. Requirements Traceability and Attributes 6
   7.1 Traceability Approach 6
   7.2 Sample RTM 6
   7.3 Requirement Attributes 6
   7.4 Acceptance Criteria Example 6
8. Tools and Techniques 7
9. Roles and Responsibilities 7
   9.1. Integration Group (Central Authority) 7
   9.2. Communication Module Team (Your Group) 7
10. Requirements Validation and Verification 8
11. Conclusion 8

1. Introduction

1.1 Purpose

This document defines the Requirements Management Plan for the **Communication Module** of the National Freelance & Skill Verification Platform. It describes the processes for planning, tracking, analyzing, prioritizing, and managing requirements throughout the project lifecycle.

1.2 Scope

The Communication Module enables secure and efficient interaction between freelancers, clients, and administrators. It includes features such as:

- Real-time messaging
- Multimedia management (file, images, videos sharing)
- Generate links for online meetings (zoom or google meet)
- One to one communication and Group chats
- Message status indicators (sent/read)
- Message Storage
- Basic Communication Interface Support
- Notification of received messages
- API/Service for Messaging
- End to end encryption of messages

This plan ensures that all requirements related to this module are systematically handled to maintain quality, traceability, and alignment with project objectives.

1.3 Definitions, Acronyms, and Abbreviations

- **SRS** – Software Requirements Specification
- **RTM** – Requirements Traceability Matrix
- **FR** – Functional Requirement
- **NFR** – Non-Functional Requirement

2. Requirements Management Approach

2.1 Requirements Planning

Requirements will be elicited using:

- Stakeholder consultations
- System analysis and benchmarking
- Functional decomposition

All requirements will be documented in the SRS and maintained in a centralized repository

2.2 Requirements Tracking

Each requirement will be assigned a unique identifier and tracked through its lifecycle. Status categories include:

- Proposed
- Approved
- In Development
- Tested
- Completed

Requirement Lifecycle:

Proposed -> Reviewed -> Approved -> In Development -> Tested -> Completed

Tracking will be performed using project management tools such as Jira or Trello.

2.3 Requirements Reporting

Progress will be monitored and reported via:

- Weekly progress reports
- Sprint review meetings

3. Configuration Management

3.1 Version Control

All requirement documents will be stored in a version-controlled repository using Git. Version history will be maintained for traceability.

3.2 Change Management Process

Changes to requirements will follow a structured process:

1. Submission of Change Request
2. Impact Analysis
3. Review and Approval
4. Implementation

5. Documentation Update

Impact Analysis will evaluate:

- Cost impact
- Schedule delay
- Dependencies with other modules
- Risk introduced

3.3 Baseline Management

- Initial baseline established after requirement approval
- Updates result in new baseline versions

3.4 Access Control

Only authorized personnel will have permission to modify requirement documents. Access will be managed using role-based permissions.

4. Requirements Prioritization

The **MoSCoW prioritization technique** will be used:

4.1 Must Have-**High**

- Real-time messaging
- Notifications system
- Message history
- Secure communication
- Group chat

4.2 Should Have-**Medium**

- Media sharing
- Meeting link

4.3 Could Have-**Low**

- Voice message
- Notification preference

4.4 Won’t Have

- AI-based chat enhancements
- Voice/Video Calling

Prioritization is based on user impact, system dependencies, and project constraints.

5.Requirements Elicitation Techniques

- Interviews
- Surveys
- Use case analysis
- Market research (existing platforms like Fiverr/Upwork)

6. Product Metrics

6.1 Performance Metrics

- Message delivery time (< 2 seconds)
- System availability (≥ 99.9%)
- Notification success rate

6.2 Quality Metrics

- Number of defects per feature
- Bug resolution time
- User feedback ratings

6.3 Usage Metrics

- Daily active users
- Number of messages exchanged
- File sharing frequency

6.4 Security Metrics

- Unauthorized access attempts
- Data encryption compliance

7. Requirements Traceability and Attributes

7.1 Traceability Approach

A Requirements Traceability Matrix (RTM) will be maintained to link:

- Requirements to design components
- Requirements to implementation tasks
- Requirements to test cases

Forward Traceability: Requirement then Design then Code then Test
Backward Traceability: Test then Requirement

7.2 Sample RTM

Requirement ID	Name	Category	Source	Status
CM-FR-01	Send Text Messages	Functional	SRS Document	In Progress – basic sending implemented
CM-FR-02	Real-Time Messaging	Functional	SRS Document	In Progress – WebSocket integration ongoing

7.3 Requirement Attributes

Each requirement will include:

- Unique ID
- Description
- Type (FR/NFR)
- Priority
- Status
- Source
- Dependencies
- Acceptance Criteria
- Version

7.4 Acceptance Criteria Example

Example:

Requirement ID: CM-FR-02
Requirement: Real-time messaging

Acceptance Criteria:

- Messages must be delivered within 2 seconds
- Messages must be stored in the database
- Messages must be visible to both sender and receiver
- Message timestamps must be recorded accurately

8. Tools and Techniques

- **Documentation:** SRS
- **Tracking:** Jira / Trello
- **Version Control:** GitHub
- **Testing:** Unit and integration testing tools

9. Roles and Responsibilities

9.1. Integration Group (Central Authority)

- Defines overall system architecture and standards
- Maintains **global requirements baseline**
- Ensures compatibility between modules
- Reviews inter-module dependencies (e.g., Communication ↔ Payment, Collaboration)
- Approves major requirement changes affecting multiple modules

9.2. Communication Module Team

9.2.1 Module Lead / Team Leader

- Manages Communication Module development
- Coordinates with Integration Group
- Ensures requirements alignment with system goals
- Approves internal requirement changes

9.2.2 Requirements Engineer / Analyst

- Gathers and documents Communication Module requirements

- Maintains SRS and RTM for this module
- Ensures clarity and completeness of requirements

9.2.3 Developers

- Implement communication features (chat, notifications, file sharing)
- Ensure requirements are correctly translated into code
- Report technical constraints

9.2.4 QA/Testers

- Validate module requirements through testing
- Develop test cases linked to requirements
- Ensure acceptance criteria are met

10. Requirements Validation and Verification

Validation:

- Requirements reviewed with stakeholders
- Ensures correctness, completeness, consistency

Verification:

- Test cases mapped to requirements
- Acceptance criteria checked during testing

11. Conclusion

This Requirements Management Plan establishes a structured framework for managing the requirements of the Communication Module. By incorporating systematic planning, tracking, prioritization, configuration control, and performance measurement, the project ensures the development of a secure, scalable, and user-friendly communication system. The adoption of these practices will minimize risks, enhance collaboration among stakeholders, and contribute to the successful implementation of the overall platform.